

Workflow: From an elevation model to a terrain map

project("EPSG:xxxxx")

every terrain variable is a change in elevation over a distance on the ground, so the raster goes into a projected CRS before anything is calculated from it

then

terrain(v = , unit =)

slope, aspect, TPI, TRI, roughness and flowdir, one layer for each named in v; aspect is circular, so take cos() of it in radians rather than its mean

then

How do you want to show the surface?

on a flat map

shade(slope, aspect, angle, direction)

slope and aspect in radians; angle is the elevation of the light and direction its bearing, conventionally 315, because a light from the southeast inverts the relief

then

ggnewscale::new_scale_fill()

sits between two geom_spatraster() calls: the grey hillshade below, the colored layer above it with alpha, so the relief shows through the colors

in three dimensions

rayshader::raster_to_matrix()

rayshader works on a matrix rather than a SpatRaster; the matrix is the transpose of the raster's grid

then

sphere_shade() ray_shade() add_shadow()

then plot_map() for an image or plot_3d() for a scene; zscale is the horizontal cell spacing divided by the vertical unit, and ray_shade() is slow: build it small